

Investigating the effect of Valerenic acid on inhibiting calcium deposition in vascular smooth muscle cells

CATEGORY: CELLULAR AND MOLECULAR BIOLOGY

BACKGROUND

Atherosclerosis: Plaque consisting of cholesterol, fat, and cellular waste build on arterial walls.

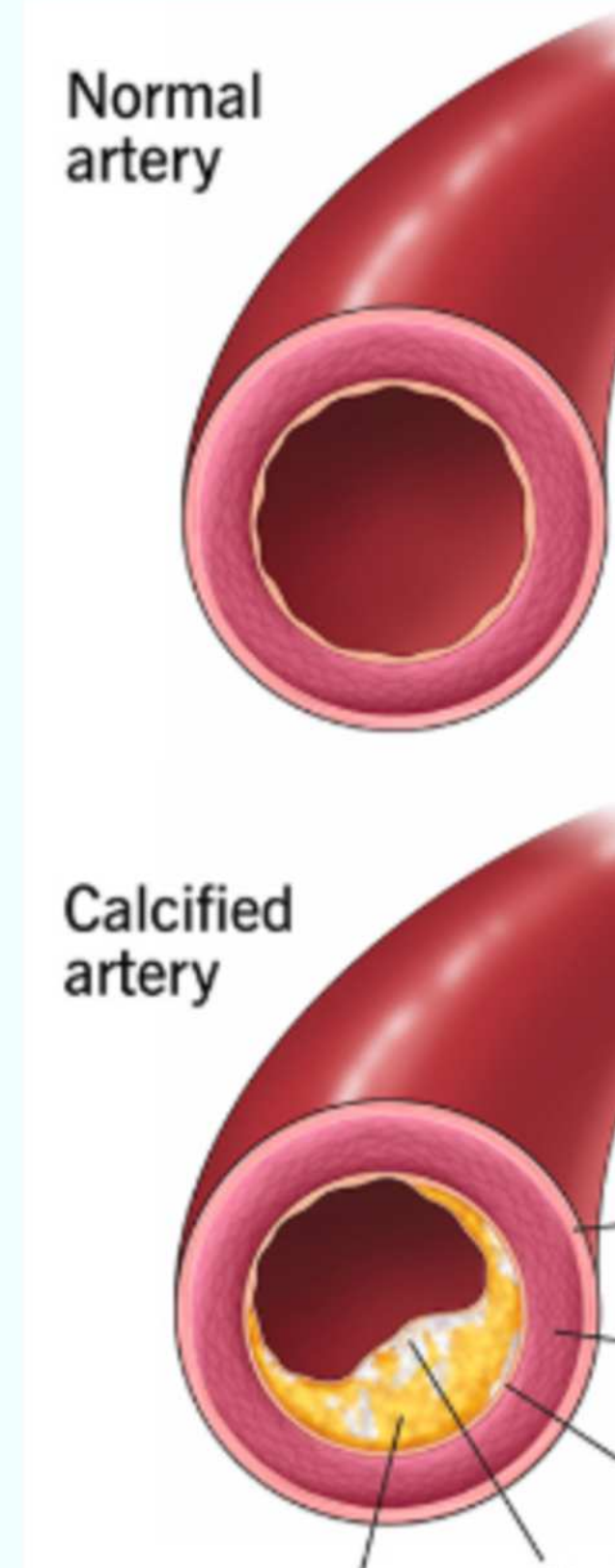
Over time, it narrows the arteries, restricting blood flow and oxygen to vital organs.

Cause: Age + Diet

Generalized Arterial Calcification of Infancy (GACI):

Rare genetic disorder. It causes abnormal calcium buildup in the walls of arteries and major blood vessels before birth or in early infancy, often leading to heart failure, stroke, and severe hypertension.

Cause: Mutation in ENPP1 and ABCC6



SIGNIFICANCE

- GRT - Expensive and in development
- Valerenic Acid would be a natural + affordable option
- Valerian Root is anti-inflammatory and is an alternative to melatonin which has been supported to inhibit calcification

HYPOTHESIS

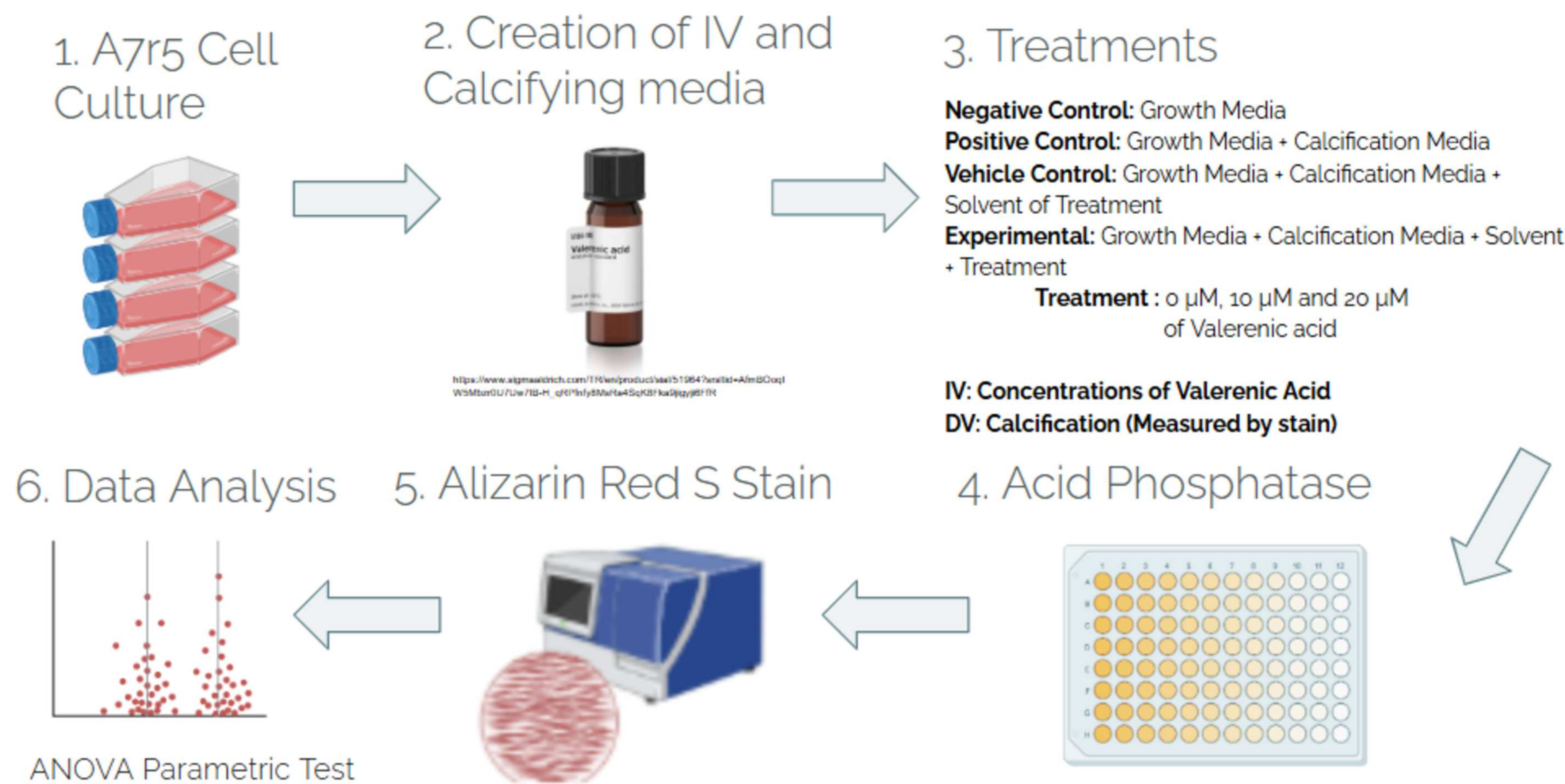
If:

Calcified VSMC's (A7r5) are treated with Valerenic Acid

Then:

Calcification will be inhibited, as measured by Alizarin Red S Staining

METHODOLOGY



DISCUSSION

Calcification of A7r5 Cells were completed, Treatment solution of Valerenic Acid was created

- IV has yet to be tested to inhibit vascular calcification
- Acid Phosphatase, Alizarin Red S Stain, and statistical analysis has not yet been completed.

IMPLICATIONS

- Valerenic acid may be used as a natural treatment for atherosclerosis and GACI - Calcium inhibition
- Provide an IV for future researchers to explore
- Provides more information on calcification of Vascular Smooth Muscle Cells

NEXT STEPS

- Run a western blot to determine anti-inflammatory proteins
- Complete Stain and Acid phosphatase to determine results of experiment

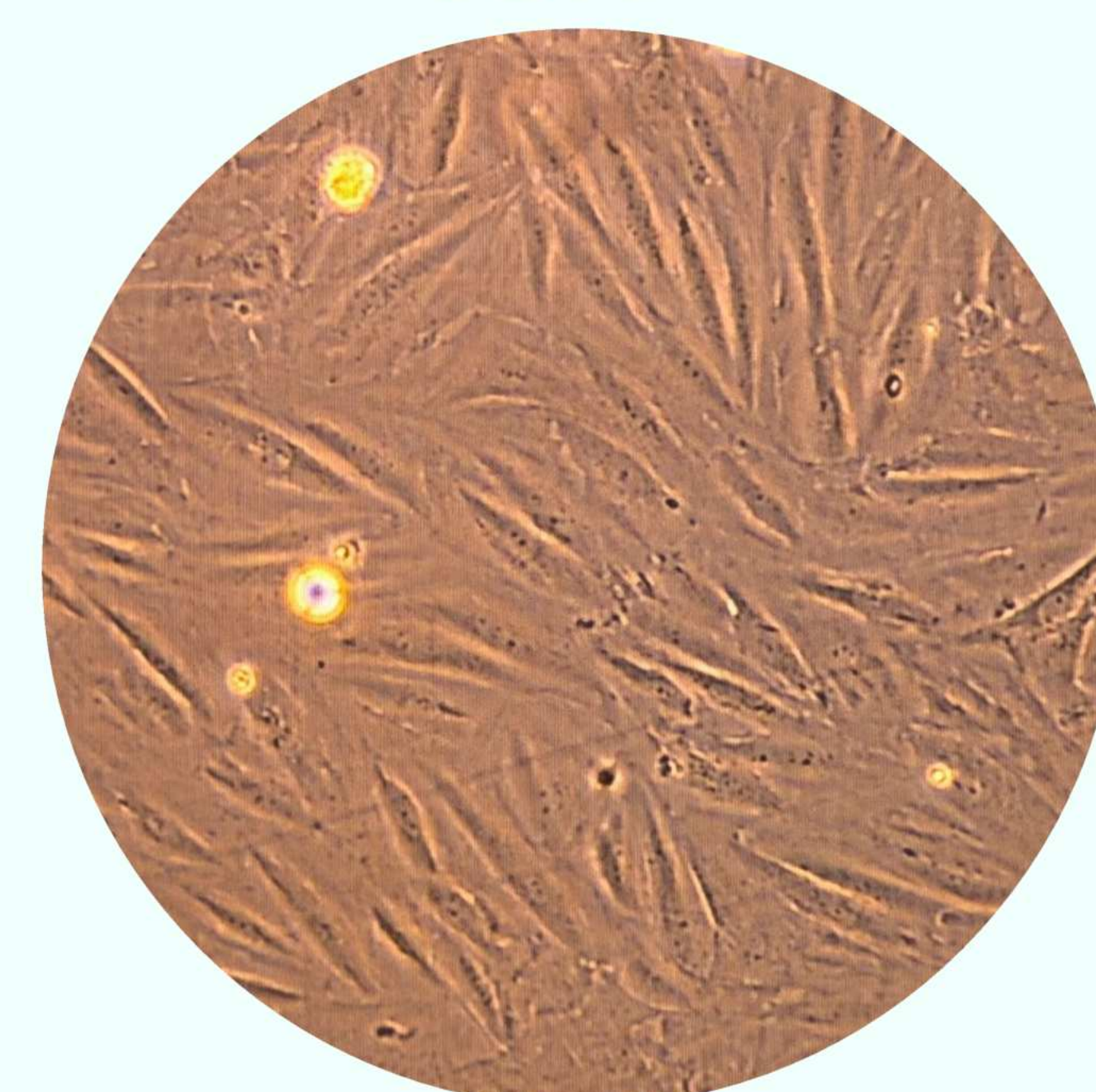
REFERENCES

- Atherosclerosis: Symptoms, Causes & Treatment. (n.d.). Cleveland Clinic. Retrieved April 22, 2025, from <https://my.clevelandclinic.org/health/diseases/16753-atherosclerosis-arterial-disease>
- Bolster, F., Ali, Z., Southall, P., Fowler, D. I. (2015). Generalized arterial calcification of infancy—Findings at post-mortem computed tomography and autopsy. *Forensic Science International*, 254, e7–e12. <https://doi.org/10.1016/j.forsciint.2015.06.017>
- Chen, W. R., Zhou, Y. J., Yang, J. Q., Liu, F., Wu, X. P., & Sha, Y. (2020). Melatonin Attenuates Calcium Deposition from Vascular Smooth Muscle Cells by Activating Mitochondrial Fusion and Mitophagy via an AMPK/OPA1 Signaling Pathway. *Oxidative medicine and cellular longevity*, 2020, 5298483. <https://doi.org/10.1155/2020/5298483>
- Coronary Artery Calcification: Causes, Symptoms & Treatment. (n.d.). Cleveland Clinic. Retrieved April 22, 2025, from <https://my.clevelandclinic.org/health/diseases/22953-coronary-artery-calcification>
- Dabravolski, S. A., Sukhorukov, V. N., Melnichenko, A. A., Khotina, V. A., & Orekhov, A. N. (2023). Potential Application of the Plant-Derived Essential Oils for Atherosclerosis Treatment: Molecular Mechanisms and Therapeutic Potential. *Molecules* (Basel, Switzerland), 28(15), 5673. <https://doi.org/10.3390/molecules28155673>
- Evans, R. (2018). Essential Oils for Heart Health: What You Need to Know. *Healthline*. <https://www.healthline.com/health/heart-disease/essential-oils-for-heart-health>
- Ferreira, C. R., Hackbarth, M. E., Ziegler, S. G., Pan, K. S., Roberts, M. S., Rosing, D. R., Whelpley, M. S., Bryant, J. C., Macnamara, E. F., Wang, S., Müller, K., Hartley, I. R., Chew, E. Y., Corden, T. E., Jacobsen, C. M., Holm, I. A., Rutsch, F., Dikoglu, E., Chen, M. Y., Mughal, M. Z., ... Gahl, W. A. (2021). Prospective phenotyping of long-term survivors of generalized arterial calcification of infancy (GACI). *Genetics in medicine: official journal of the American College of Medical Genetics*, 23(2), 396–407. <https://doi.org/10.1038/s41436-020-00983-0>
- GACI Global. (n.d.). Why is it important to have an awareness day for GACI/ARH2? GACI Global. <https://gaciglobal.org/awareness-day/#:~:text=GACI%2FARH2%20are%20rare,GACI%20do%20not%20survive%20infancy>
- Pahwa R, Jialal I. Atherosclerosis. (2025). In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. <https://www.ncbi.nlm.nih.gov/books/NBK507799/>
- Ralph, D., Nitschke, Y., Levine, M. A., Caffet, M., Wurst, T., Saeidian, A. H., Youssefian, L., Vahidnezhad, H., Terry, S. F., Rutsch, F., Uitto, J., & Li, Q. (2022). ENPP1 variants in patients with GACI and PXE expand the clinical and genetic heterogeneity of heritable disorders of ectopic calcification. *PLoS genetics*, 18(4), e1010192. <https://doi.org/10.1371/journal.pgen.1010192>

Data Progress/Visualization

BEFORE EXPOSURE

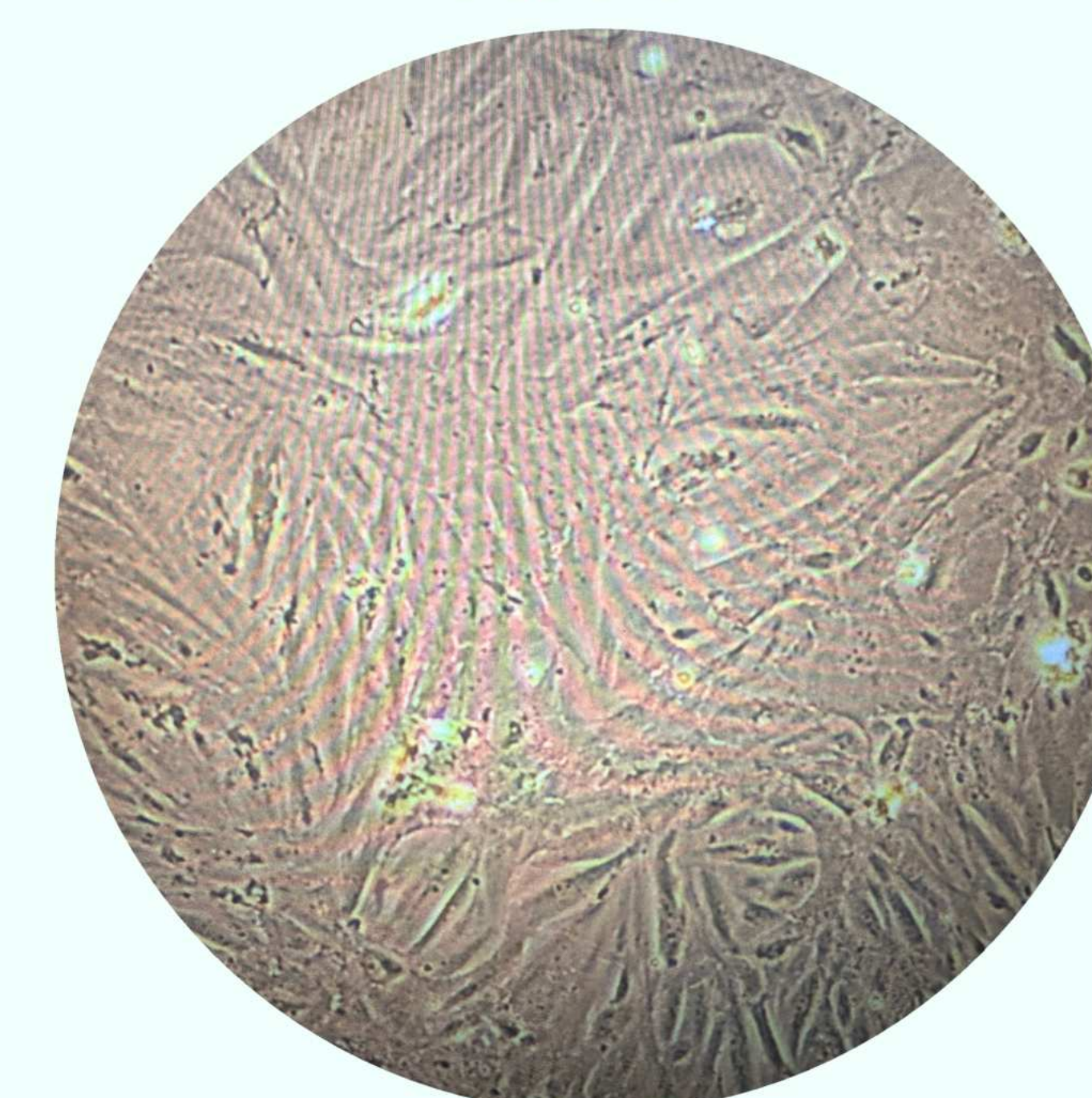
A7r5



100x

AFTER EXPOSURE

A7r5



100x

Transformation Observations

- Accumulations/Buildups of calcium
- Dystrophic calcification
- Slowed cellular proliferation
- Increase in visibility of extracellular vesicles

Calcification of VSMC's

- 1 week of exposure to calcification media
- The media includes DMEM, FBS, CaCl₂, and a sodium phosphate buffer.
- Replicates conditions of atherosclerosis and GACI